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Large Language Models to Analyze Business Intelligence User Narratives

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ABSTRACT

Enterprises increasingly recognize the value of Business Intelligence (BI) systems in enabling data-informed decision-making. However, prior studies using structured data, surveys, and case studies often lack the granularity to capture deployment dynamics across diverse organizational settings. This study addresses that gap by analyzing customer testimonials from eight leading BI platforms. We propose a Large Language Model (LLM)-based framework for systematically processing and extracting insights from unstructured narratives. Using GPT-4, we analyzed approximately 2800 testimonials scraped from vendor websites to identify adoption drivers, implementation strategies, challenges, collaboration dynamics, and cross-platform trends. Our approach enables large-scale, structured insight extraction, transforming fragmented narratives into organized knowledge that informs both research and practice. The analysis also reflects how user perspectives and vendor framings shape BI platform narratives. This study demonstrates the potential of LLMs for scalable, automated qualitative analysis and contributes to understanding how generative AI can uncover actionable insights from unstructured enterprise data.

KEYWORDS

Business intelligence; large language models; text mining; customer narratives; generative AI

Introduction

In the era of big data analytics, businesses are increasingly turning to business intelligence (BI) systems to transform raw data into actionable insights, driving strategic decision-making and competitive advantage.[1–3] The widespread use of platforms such as Power BI, Tableau, and Looker underscores the need to understand how these tools are implemented across diverse business contexts. While technical guides and theoretical models exist, empirical evidence is needed to identify the practical factors that distinguish successful BI initiatives.

BI systems are technology-driven processes used by organizations to analyze data and present actionable information.[4,5] The BI landscape is evolving rapidly, with modern systems incorporating advanced data visualization, real-time analytics, and AI/ML integration. These developments enhance data interaction, enable prompt responses to market changes, and automate analysis for predictive insights. Understanding how organizations leverage these capabilities is crucial for achieving business objectives.[6–8]

Key motivations for using BI include gaining a competitive advantage,[9,10] enhancing customer service,[10] optimizing resource allocation,[11] and uncovering new business opportunities.[2] From a technological perspective, factors such as the perceived relative advantage of BI, its compatibility with existing systems, complexity, trialability, and observability have been identified as significant influences.[12]

However, the literature also reveals common challenges that can hinder successful BI implementation. These include issues related to data quality and integration, lack of skilled personnel, resistance to change, and misalignment between BI capabilities and business goals.[2,13,14]

Although numerous technical guides and theoretical models have been developed,[15,16] they often focus primarily on theoretical benefits and technical aspects, with limited evidence from actual practice. This gap highlights the need for an investigation into real-world, practical experiences. Customer testimonials, often found in the 'Customer Stories' sections of BI vendor websites, present a valuable resource in this regard. These testimonials provide first-hand accounts of the motivations behind BI initiatives, the triumphs and challenges encountered during implementation, and the long-term strategic impact of data-driven insights.

However, extracting meaningful patterns and actionable recommendations from customer testimonials poses a distinct challenge. Their largely unstructured

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nature, filled with narratives and individual experiences, makes it difficult to apply traditional quantitative analysis methods. This highlights the need for a methodology that systematically processes and interprets the rich qualitative data embedded in these testimonials, maximizing their potential to guide successful BI implementations. Recent research has emphasized the potential of integrating generative artificial intelligence (Gen AI) to enhance the analysis of qualitative data in literature reviews.[17–19]

Large language models (LLMs)[20–23] are advanced artificial intelligence systems trained on large-scale text datasets. By understanding the complexities and patterns of human language, these models have demonstrated strong performance across a range of natural language processing (NLP) tasks, including language translation, answering questions, and summarizing text. Recent research has highlighted significant progress made by LLMs, revealing substantial improvements across various NLP tasks and benchmarks.[24–27]

This study introduces a novel application of LLMs to analyze customer testimonials published by BI platform vendors. The aim is to identify the practical factors shaping BI adoption and implementation by examining motivations, strategies, challenges, and the organizational outcomes reported by users. Rather than modeling adoption behavior, we focus on uncovering patterns from authentic organizational accounts using an LLM-based analysis pipeline. This information can be instrumental in guiding successful BI deployments by providing a deeper understanding of the practical considerations and potential pitfalls involved in leveraging BI for business advantage. This study aims to achieve the following research objectives: (RO)

- RO1: Identify the primary motivations and strategic objectives driving BI adoption across different industries.
- RO2: Determine the most effective implementation strategies and methodologies for successful BI deployment and integration, considering industry-specific best practices.
- RO3: Analyze the common challenges encountered during BI implementation and assess their variations across different industries.
- RO4: Investigate how the use of BI systems influences organizational culture and collaboration, examining industry-specific nuances in developing a data-driven culture.

By systematically analyzing customer testimonials with the proposed methodology, this research seeks to provide empirical insights that address these objectives. This approach not only enriches the understanding of the functionalities and impacts of the BI tool, but also highlights how different companies tailor these tools to meet specific business needs and challenges.

The significance of this study lies in its potential to bridge the gap between theoretical models and the practical realities faced by businesses. By focusing on customer testimonials, we capture the experiences of organizations, providing a more nuanced understanding of the factors that contribute to successful BI implementation. This approach enriches the existing literature whilst offering practical guidance to businesses seeking to navigate the complex landscape of BI technologies. This research makes two core contributions. First, methodologically, we demonstrate how LLMs can support large-scale qualitative analysis by systematically processing rich textual data and extracting key themes from unstructured sources. Second, empirically, we surface cross-platform insights that reveal implementation patterns not typically visible through traditional research approaches. This innovative methodology enables the extraction of actionable insights from extensive qualitative datasets, offering tailored understanding that addresses the specific needs and objectives of diverse organizations.

The remainder of the paper is organized as follows: Section 2 reviews the BI literature and identifies research gaps; Section 3 explains the methodology; Section 4 presents the findings; Section 5 discusses implications for practice; and Section 6 summarizes the contributions, limitations, and future research directions.

Background

The evolution of BI systems reflects rapid advancements in information technology. Initially designed for basic data storage and retrieval, BI systems have expanded to include aggregation, descriptive analytics, and Online Analytical Processing, shifting from static reporting to interactive, real-time decision support.[28] The integration of AI and machine learning has enabled predictive capabilities, allowing organizations to forecast trends and align insights with strategic objectives.[8,29]

Organizations increasingly turn to BI systems to manage the growing complexity of today's business environments. This shift has been accelerated by advances in big data, cloud computing, and artificial intelligence, alongside declining data storage costs.[30–32] As a result, BI platforms have become central to the development of metrics-driven management practices, supporting more informed and strategic decision-

making.[10,33,34] The effective use of these systems depends on a range of factors, including organizational readiness, top management support, technological maturity, data quality, and vendor-related considerations.[35]

Despite the benefits, BI implementation also presents challenges, including data integration issues, skills shortages, user resistance, and concerns around security and privacy.[36] Organizational resistance often stems from misconceptions about BI benefits and the disruptive nature of new technologies.[37] Successful implementation requires aligning BI strategies with organizational goals and investing in skilled personnel.[38]

Resource constraints further complicate BI deployment. High costs associated with licensing, infrastructure, and maintenance can be prohibitive, particularly for smaller organizations.[39,40] The process of selecting, implementing, and integrating BI systems is resource-intensive, requiring careful planning and sustained commitment.[41,42]

A range of studies have examined BI adoption and implementation across sectors. Table 1 summarizes selected contributions that use methodologies such as systematic literature reviews, surveys, mixed-methods approaches, and conceptual frameworks.

In the literature, the adoption and implementation of BI systems have been examined using a range of approaches, including theoretical frameworks such as the Technology—Organization—Environment (TOE) model, conceptual models, and data-driven analyses.[29] Common themes include relative advantage, compatibility, complexity, top management support, and organizational readiness. Research spans domains including healthcare,[12,43] insurance,[44] higher education,[35] and government,[45] highlighting sector-specific challenges and adoption patterns.

A substantial body of work addresses BI deployment in developing countries,[35,44,46] where challenges such as infrastructure limitations and resource scarcity are common. Studies focused on Small and Medium Enterprises highlight additional issues, including financial constraints and limited technical capacity.[12,46,47] Across contexts, recurring barriers include under-utilization,[2] limited user training and communication,[45] and data quality concerns.[48]

Overall, the literature demonstrates the multifaceted nature of BI adoption and implementation, as well as the importance of context-specific, and user-centered approaches.

Online customer testimonial as a data source

Customer testimonials are a rich source of insights, offering detailed accounts of organizational

Table 1. Summary of key studies.

Reference	Industry/ Context	Methodologies	Key Findings/Contributions
Ain et al.[2]	BI Systems General Review	Systematic Literature Review	Highlighted underutilization and user resistance as key challenges; identified a lack of user-centered research.
Shaikh et al.[28]	SMEs General Review	Literature Review	Cost, lack of technical skills, adoption challenges, and lack of organizational and government support are significant challenges.
Fakier[29]	Government (South Africa)	Mixed Methods	Identified lack of communication, leadership, training, and limited awareness as key barriers.
Salisu et al.[12]	Healthcare (SMEs)	Systematic Literature Review	Identified 16 significant factors in four dimensions: technology, Organization, environment, and owner-manager characteristics
Hmoud et al.[30]	Higher Education (Jordan)	Quantitative/Partial Least Squares Structural Equation Modeling (PLS-SEM)	Information culture, complexity, relative advantage, top management support, compatibility, organizational readiness, information quality, and vendor selection are significant determinants.
El Malki and Touate[31]	Various	Systematic Literature Review	Identified over 70 TOE factors and 24 human factors;
Horani et al.[32]	Business Analytics General	Systematic Literature Review	organizational, technological, and environmental factors play significant roles in BI adoption.
Alsibhawi et al.[33]	SMEs (Libya)	Conceptual Framework	Proposed factors include change management, knowledge sharing, information quality, IT project management, perceived usefulness, and ease of adoption.
Alzaabi et al.[34]	Healthcare (UAE)	Cross-Sectional Survey	Cultural values don't moderate adoption; trust and expectancy are key factors.
Solanki et al.[35]	Various/ General Business	Literature Review	Emphasized the role of BI in data-driven decision-making and operational efficiency.
Jaradat et al.[36]	Insurance (Jordan)	Quantitative/PLS-PM	Performance expectancy, social influence, facilitating conditions, and perceived trust predict BI adoption intention; cultural values did not significantly moderate relationships.

motivations, experiences, and perceived outcomes. While several studies have analyzed testimonials,[49–52] systematic text-mining approaches remain limited. Prior research using NLP has shown potential,[53] yet is often constrained by small sample sizes (typically 50–200 stories) and reliance on manual coding, which introduces subjectivity and limits generalizability.

Generative AI and LLMs offer an opportunity to analyze larger testimonial datasets with greater consistency. They can support text extraction, summarization, and classification, reducing human bias and enabling scalable analysis. Their ability to process unstructured narratives offers a more efficient route to identifying themes in customer experiences.

However, testimonials curated by vendors inherently present positive cases, limiting the presence of neutral or negative experiences. This selective presentation inherently skews the analysis toward favorable outcomes. While this restricts their use in evaluating BI effectiveness, testimonials remain valuable for understanding how organizations articulate their motivations, challenges, and the strategic use of BI tools

Research gap and opportunities

Existing BI research has advanced the understanding of the technological, organizational, and environmental determinants of adoption. However, most empirical studies rely on surveys with relatively small samples, limiting their ability to capture the diverse realities of BI implementation. To address this gap, our study analyses 2,800 vendor-published customer testimonials, offering a significantly broader dataset that reflects cross-industry experiences.

Although factors such as infrastructure, management support, and organizational readiness are widely recognized as critical enablers of BI adoption,[35,44] user-driven perspectives remain underexplored. Testimonials represent an important narrative source through which organizations construct success stories, describe challenges, and articulate perceived value. Understanding these perspectives provides an opportunity to extend prevailing frameworks and identify overlooked dynamics.

This study addresses four gaps: (1) The need for comprehensive analyses of large-scale customer narratives; (2) Limited insight into user-centric success narratives and pain points; (3) Insufficient understanding of how challenges are framed in real-world accounts; (4) Underutilization of AI and NLP for analyzing unstructured customer testimonials data.

Rather than modeling behavior, this study extracts patterns from customer experiences using LLM-based analysis. Our approach advances BI research by applying an AI-driven methodology to analyze large-scale testimonials. Our findings demonstrate how large-scale real-world narratives can be systematically analyzed to reveal implementation dynamics across industries, offering a scalable, data-driven perspective on BI deployment.

By focusing on textual data from BI platforms, our research extracts themes such as adoption drivers, implementation strategies, and challenges, offering enterprise-tailored perspectives. This customer-centric approach addresses the current gap by not only identifying emerging trends and challenges but also proposing actionable insights derived from user experiences.

Methodology

This study adopts a multi-stage methodological design combining a large-scale collection of customer testimonials, LLM-based extraction of predefined variables, and the consolidation of outputs for statistical analysis. Figure 1 summarizes the overall workflow adopted in this study.

BI platform selection

Selecting appropriate BI platforms for evaluation is a crucial aspect of this research. Drawing on the findings of the literature review,[5,54,55] we selected BI platforms widely recognized and adopted across the industry. The selection reflects their established standing in the Gartner Magic Quadrant recognition[55] and their broad use across both enterprise and SME settings. These platforms offer comprehensive BI functionalities, including data integration, analysis, visualization, and reporting. Additionally, we ensured that these platforms have accessible online customer success stories to facilitate our data collection. The platforms included in this study are Board, Dash, Insight, KNIME, Looker, Power BI, Qlik, and Tableau.

Although categorized as a "Data Science and Machine Learning Platform,"[55] KNIME is included due to substantial overlap with BI functionality. Its data integration, transformation, analysis, and visualization capabilities align with BI systems' operational scope. KNIME's inclusion captures the convergence of traditional BI and advanced analytics in current BI practices.

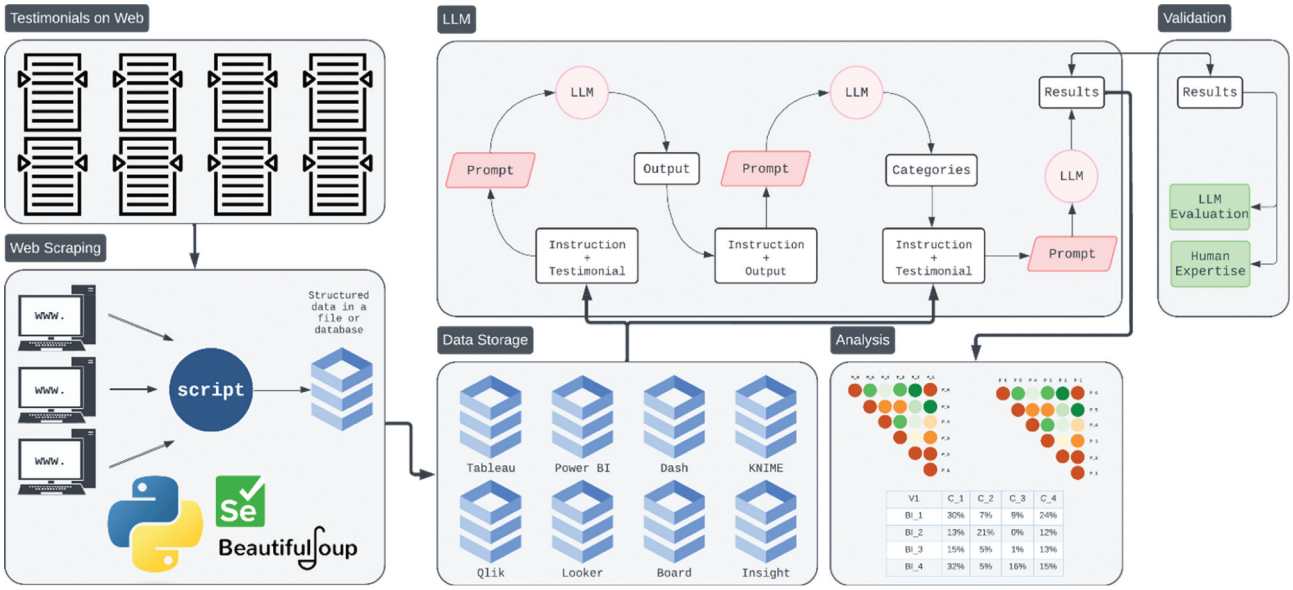


Figure 1. Research methodology.

Customer testimonial collection

Testimonials were collected by scraping customer story pages from the eight platforms. Each script was adapted to the structure of the respective website. Challenges arising from dynamic content and inconsistent HTML structures were addressed through delay mechanisms, error handling, and targeted parsing logic.

BeautifulSoup and Selenium in Python[56,57] were used for parsing and automated navigation. Scripts located the "Customer Stories" sections and extracted testimonial text, titles, URLs, and company names. Table 2 reports the number of stories collected per platform and their average word counts. In total, 2,865 testimonials were gathered. The text was preprocessed through HTML removal, lower-casing, and stopword filtering. Stories that were videos, irrelevant, or too short were excluded.

Industry classification

Industry sectors were assigned using a combined UK SIC classification[58] and LLM-based categorization procedure. Using story content, the model identified three plausible industries, with the top choice retained. Sector labels were used to structure subsequent analysis.

To assess accuracy, classifications were compared with a dataset of approximately 17 million companies containing information on industry, size, and funding.[59] Because the dataset's categories differed from those in the UK SIC, a manual mapping step was used to ensure alignment. Manual inspection indicated model accuracy above 90%; remaining cases typically showed more than one plausible category. For example, UK SIC "70.22: Business and management consultancy activities" aligns with the broader category "Professional Services." This ensured consistency when comparing the external dataset with UK SIC categories.

The resulting distribution shows a concentration of testimonials in a small number of sectors: professional, scientific and technical activities; information and communication; financial and insurance activities; and wholesale and retail trade, which together account for nearly half of all stories. Table 3 reports the sectoral breakdown by platform.

Table 2. Information on collected datasets.

BI Software	# of Stories	Average Word Count
Tableau	759	766
Power BI	624	1288
Looker	829	943
Insight	228	663
Board	142	864
Dash	24	664
KNIME	94	697
Qlik	165	583
Total	2865	808

Selection process of variables

The selection of variables was informed by prior research on BI adoption and implementation.[10,13,16,60–62] Recurring themes were used to define four variables aligned with the research objectives:

- Adoption Drivers: motivations and strategic objectives driving BI adoption (e.g. operational

Table 3. Breakdown of industry sectors and stories by bi platforms.

Industry Sector	Board freq. (%)	Dash freq. (%)	Insight freq. (%)	KNIME freq. (%)	Looker freq. (%)	Power BI freq. (%)	Qlik freq. (%)	Tableau freq. (%)
Accommodation And Food Service Activities (106)	20 (16)	1 (4)	8 (4)	0 (0)	22 (3)	43 (7)	2 (8)	10 (1)
Administrative And Support Service Activities (120)	10 (8)	0 (0)	6 (3)	3 (3)	19 (2)	32 (5)	0 (0)	50 (7)
Agriculture, Forestry and Fishing (17)	4 (3)	0 (0)	3 (2)	0 (0)	0 (0)	8 (1)	0 (0)	2 (0)
Arts, Entertainment and Recreation (51)	3 (2)	1 (4)	3 (2)	0 (0)	15 (2)	9 (1)	1 (4)	19 (3)
Construction (97)	4 (3)	1 (4)	18 (10)	9 (10)	19 (2)	33 (5)	0 (0)	13 (2)
Education (91)	2 (2)	1 (4)	1 (1)	1 (1)	10 (1)	30 (5)	1 (4)	45 (6)
Electricity, Gas, Steam and Air Conditioning Supply (86)	8 (6)	5 (21)	10 (5)	0 (0)	6 (1)	40 (6)	2 (8)	15 (2)
Financial And Insurance Activities (299)	13 (10)	2 (8)	25 (13)	13 (15)	96 (12)	66 (11)	3 (12)	81 (11)
Human Health and Social Work Activities (201)	8 (6)	2 (8)	19 (10)	8 (9)	39 (5)	53 (9)	0 (0)	72 (10)
Information and Communication (417)	7 (6)	5 (21)	19 (10)	16 (19)	217 (27)	46 (7)	3 (12)	104 (15)
Manufacturing (141)	16 (13)	2 (8)	28 (15)	5 (6)	12 (1)	44 (7)	4 (15)	30 (4)
Mining and Quarrying (5)	0 (0)	0 (0)	1 (1)	0 (0)	1 (0)	1 (0)	0 (0)	2 (0)
Other (10)	0 (0)	0 (0)	3 (2)	0 (0)	2 (0)	1 (0)	0 (0)	4 (1)
Other Service Activities (16)	2 (2)	0 (0)	1 (1)	1 (1)	1 (0)	1 (0)	0 (0)	10 (1)
Professional, Scientific and Technical Activities (478)	3 (2)	2 (8)	19 (10)	20 (23)	213 (27)	89 (14)	3 (12)	129 (18)
Public Administration and Defense; Compulsory Social Security (38)	1 (1)	0 (0)	2 (1)	2 (2)	3 (0)	18 (3)	0 (0)	12 (2)
Real Estate Activities (30)	0 (0)	0 (0)	6 (3)	0 (0)	7 (1)	8 (1)	0 (0)	9 (1)
Transportation and Storage (72)	5 (4)	2 (8)	7 (4)	0 (0)	7 (1)	29 (5)	1 (4)	21 (3)
Water Supply; Sewerage, Waste Management and Remediation Activities (11)	0 (0)	0 (0)	0 (0)	0 (0)	1 (0)	9 (1)	0 (0)	1 (0)
Wholesale and Retail Trade (297)	20 (16)	0 (0)	10 (5)	8 (9)	113 (14)	63 (10)	6 (23)	77 (11)

efficiency, competitive advantage, improved decision-making).

- Implementation Strategies: approaches and methodologies used to deploy and integrate BI in the organization.
- Challenges and Solutions: obstacles encountered during implementation and the responses adopted.
- Collaboration and Culture: initiatives and practices that support a data-driven culture, collaboration and data-informed decision-making.

LLM-based analysis

The methodology employed in this study utilizes OpenAI's GPT-4,[63] a LLM known for its enhanced capabilities in natural language understanding and generation. GPT-4 features significantly more parameters than its predecessors, enabling enhanced performance across a variety of tasks.[64,65]

GPT-4 incorporates supervised fine-tuning on human feedback and an extensive and diverse dataset, increasing its reliability and versatility across application scenarios ranging from simple text generation to complex problem-solving tasks.[66] This makes it particularly suited for analyzing and summarizing detailed customer testimonials, where nuanced understanding and contextual comprehension are key. Its advanced architecture allows for the distillation of complex, verbose inputs into concise, insightful summaries, effectively sifting through extraneous information to pinpoint critical data.[67–69]

GPT-4 was configured with a temperature of 0.7 and a maximum token limit of 3072. The temperature parameter was selected to balance creativity and conciseness in its responses, as a higher temperature increases randomness and diversity, while a lower temperature favors more deterministic outputs. The maximum token limit was determined through preliminary tests to ensure comprehensive yet coherent outputs for given prompts, mitigating the risk of either truncation or overly verbose results.

As presented in Figure 2, the analysis consisted of three cycles. In the first cycle, individual stories were fed into the model with a prompt designed to extract key insights for each predefined variable. An example JSON structure was provided to standardize the output format, and the model returned a JSON file[70] containing variable-level outputs for each story.

In the second cycle, the text associated with each variable (e.g., all implementation strategy outputs) was aggregated and reentered into the model. The prompt instructed the model to group these into a set of classes and to disregard content that did not fit. This produced an initial set of categories for each variable.

In the third cycle, the original testimonial content for each variable was re-analyzed, with the model instructed to assign one or more predefined classes. This step generated a structured dataset that linked each story to one or more categories for each variable.

To reduce hallucination, prompts included explicit instructions not to infer information where the testimonial did not mention a variable. In such cases, the model was instructed to return "information not found." In an

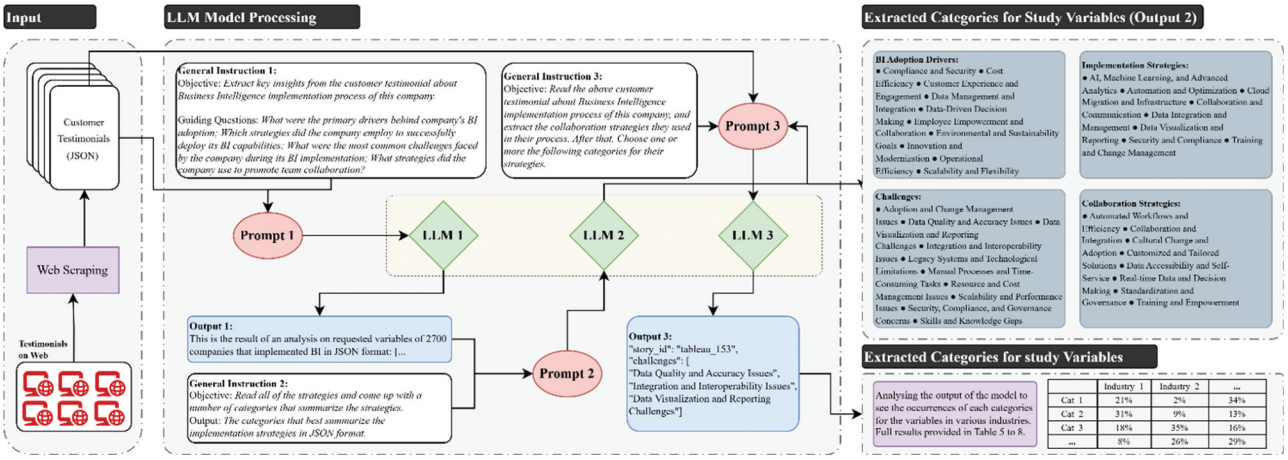


Figure 2. Conceptual framework for LLM-based analysis of customer narratives.

initial pilot on 40 stories, adding this instruction reduced the observed hallucination rate from 12% to 4%.[71,72]

Prompt design followed established principles[73] of clarity, contextualization and iterative refinement. Prompts specified the variables of interest, provided brief examples and explained the analysis context. Effectiveness was assessed on a small subset of stories and adjusted before applying the final prompts to the full dataset. Three prompts were used: one to extract variable-level insights, one to generate categories and one to assign categories to each testimonial. The exact prompts are reported in Appendix A.

To address potential bias in vendor-curated testimonials, the LLM was explicitly instructed to identify any challenges, implementation barriers or difficulties mentioned in the narratives, even when these appeared within otherwise positive accounts. This allowed the analysis to capture challenges that may be expressed subtly or alongside positive outcomes, providing a more balanced view of customers' adoption and implementation experiences.

Validation

Validating variable extraction and text classification can be difficult when dealing with unsupervised learning tasks, particularly in the absence of established benchmarks. We therefore used a multi-pronged evaluation strategy combining human inter-rater reliability (IRR) and LLM-based evaluation.[74,75]

For IRR, two independent human coders categorized a random sample of 40 testimonials across all variables using the same framework as GPT-4. Cohen's kappa[76] was used to assess agreement. Human—human agreement ranged from 0.62 to 0.78 (average $\kappa = 0.73$). Comparing GPT-4 classifications with human consensus yielded similar levels of agreement, with $\kappa = 0.74$ for adoption drivers, $\kappa = 0.72$ for implementation strategies and $\kappa = 0.63$ for collaboration.

We also conducted a comparative analysis using LLaMA-3[77] as an additional benchmark. Cohen's kappa between GPT-4 and LLaMA-3 was 0.79, indicating substantial agreement. GPT-4, however, showed advantages in terms of speed and consistency.

Finally, we manually reviewed approximately 40 testimonials across multiple GPT-4 iterations to assess coherence, accuracy and potential hallucinations. This review informed refinements to both model configuration and prompt design. Overall, the validation procedure assessed the model's ability to identify and categorize key insights within the qualitative data. Variables, categories and representative excerpts were extracted from the testimonial texts, with examples provided in Appendix Table B.9.

Findings

The findings presented in this section are categorized according to the UK SIC codes.[58] To facilitate readability, these codes are used throughout tables and figures. Through the second cycle of the methodology, the LLM distilled the extracted text for each variable into a set of categories. As shown in Figure 2, we used a hybrid approach: the LLM generated initial categories from the testimonial data, and we then checked these against themes reported in the BI literature. The final categories, summarized in Table 4, form the basis of the results.

RO1: Adoption drivers

The analysis of BI adoption drivers (Table 5) indicates a consistent emphasis on improving organizational

Table 4. Extracted categories.

Variables	Key theme	Brief Description
Adoption Drivers	Operational Efficiency	Optimize processes, automate tasks, enhance productivity.
	Data Management and Integration	Consolidate and govern data, create a unified source of truth.
	Scalability and Flexibility	Adapt BI solutions to evolving data needs and business requirements.
	Cost Efficiency	Deliver tangible cost savings and favorable ROI.
	Customer Experience and Engagement	Gain customer insights, facilitate personalized interactions.
	Data-Driven Decision Making	Transform data into actionable intelligence for informed decisions.
	Innovation and Modernization	Drive digital transformation, foster innovation culture.
	Compliance and Security	Safeguard data, adhere to regulatory standards.
	Employee Empowerment and Collaboration	Enable data access and insights at all levels.
	Environmental and Sustainability Goals	Contribute to sustainable business practices.
Implementation Strategies	Data Integration and Management	Consolidate and organize data from various sources for unified BI foundation.
	Data Visualization and Reporting	Present data through meaningful visualizations, dashboards, and reports.
	Cloud Migration and Infrastructure	Leverage cloud platforms for scalable, flexible, and cost-efficient BI deployment.
	AI, Machine Learning, and Advanced Analytics	Integrate cutting-edge technologies for deeper insights and predictive capabilities.
	Collaboration and Communication	Foster knowledge sharing and BI adoption through effective tools.
	Security and Compliance	Protect sensitive data and ensure regulatory compliance in BI environments.
	Training and Change Management	Focus on user adoption through training, support, and internal champions.
	Automation and Optimization	Streamline processes and maximize efficiency through automation techniques.
Challenges	Manual Processes and Time-Consuming Tasks	Inefficiencies from manual data handling, leading to delays and errors.
	Data Quality and Accuracy Issues	Concerns about inconsistencies, duplication, and errors in data.
	Integration and Interoperability Issues	Difficulties in integrating disparate systems and data sources.
	Data Visualization and Reporting Challenges	Struggles with creating effective visualizations and actionable reports.
	Scalability and Performance Issues	Concerns about handling increased workloads and large datasets.
	Legacy Systems and Technological Limitations	Hindrance from outdated technologies and inflexible solutions.
	Skills and Knowledge Gaps	Need for training and addressing resistance to change.
	Security, Compliance, and Governance Concerns	Challenges in ensuring data security, privacy, and regulatory compliance.
	Resource and Cost Management Issues	Strain on resources due to budget constraints and maintenance costs.
	Adoption and Change Management Issues	Overcoming cultural resistance and ensuring user adoption.
Collaboration	Training and Empowerment	Comprehensive programs to enable effective use of BI tools.
	Collaboration and Integration	Breaking down data silos and facilitating seamless teamwork.
	Data Accessibility and Self-Service	Democratizing insights through independent data exploration.
	Real-time Data and Decision Making	Empowering agile decision-making with up-to-date information.
	Standardization and Governance	Ensuring data quality, consistency, and security.
	Cultural Change and Adoption	Cultivating a data-driven mindset across the organization.
	Automated Workflows and Efficiency	Streamlining processes to focus on strategic initiatives.
	Customized and Tailored Solutions	Addressing specific user needs with personalized BI approaches.

Table 5. BI adoption drivers for different sectors.

	Industry Sector*																			
	I-1	I-2	I-3	I-4	I-5	I-6	I-7	I-8	I-9	I-10	I-11	I-12	I-13	I-14	I-15	I-16	I-17	I-18	I-19	I-20
Compliance and Security	4%	2%	0%	1%	2%	3%	3%	6%	6%	4%	3%	8%	0%	0%	4%	5%	3%	3%	4%	2%
Cost Efficiency	7%	6%	7%	6%	8%	5%	8%	7%	6%	8%	8%	12%	6%	4%	10%	7%	3%	7%	6%	7%
Customer Experience and Engagement	6%	8%	4%	12%	5%	4%	3%	8%	7%	10%	4%	0%	6%	6%	8%	5%	11%	9%	9%	11%
Data Management and Integration	15%	16%	18%	15%	14%	16%	14%	13%	14%	12%	16%	20%	15%	19%	12%	14%	15%	15%	14%	14%
Data-Driven Decision Making	16%	18%	16%	18%	16%	19%	14%	14%	16%	15%	16%	12%	19%	20%	14%	16%	17%	16%	13%	16%
Employee Empowerment and Collaboration	11%	9%	12%	10%	10%	15%	12%	9%	11%	7%	11%	4%	6%	11%	9%	10%	11%	10%	9%	8%
Environmental and Sustainability Goals	2%	1%	1%	0%	4%	1%	5%	1%	0%	1%	1%	4%	0%	0%	1%	3%	1%	1%	10%	1%
Innovation and Modernization	11%	10%	12%	10%	12%	10%	14%	12%	11%	12%	12%	16%	17%	10%	12%	13%	11%	11%	13%	12%
Operational Efficiency	17%	18%	20%	18%	18%	19%	18%	18%	17%	17%	19%	20%	21%	20%	18%	18%	19%	17%	14%	18%
Scalability and Flexibility	11%	11%	11%	11%	11%	9%	11%	12%	10%	14%	11%	4%	9%	10%	13%	9%	10%	10%	9%	13%

*Note: (I-1) Accommodation and food service activities, (I-2) Administrative and support service activities, (I-3) Agriculture, forestry and fishing, (I-4) Arts, entertainment and recreation, (I-5) Construction, (I-6) Education, (I-7) Electricity, gas, steam and air conditioning supply, (I-8) Financial and insurance activities, (I-9) Human health and social work activities, (I-10) Information and communication, (I-11) Manufacturing, (I-12) Mining and quarrying, (I-13) Other, (I-14) Other service activities, (I-15) Professional, scientific and technical activities, (I-16) Public administration and defence; compulsory social security, (I-17) Real estate activities, (I-18) Transportation and storage, (I-19) Water supply; sewerage, waste management and remediation activities, and (I-20) Wholesale and retail trade.

performance. "Operational Efficiency" is the most frequently observed driver across sectors (19%). This aligns with the broader BI literature, which positions BI as a mechanism for reducing manual processes, automating workflows, and supporting streamlined operations.[33,48]

"Data-Driven Decision Making" (17%) is similarly prominent, reflecting the widespread recognition that

Table 6. Implementation strategies for different sectors.

	Industry Sector*																			
	I-1	I-2	I-3	I-4	I-5	I-6	I-7	I-8	I-9	I-10	I-11	I-12	I-13	I-14	I-15	I-16	I-17	I-18	I-19	I-20
AI, Machine Learning, and Advanced Analytics	4%	6%	2%	9%	8%	5%	8%	11%	6%	14%	5%	12%	3%	8%	11%	5%	10%	5%	2%	9%
Automation and Optimization	9%	7%	9%	6%	10%	6%	10%	8%	7%	7%	11%	12%	3%	6%	8%	11%	7%	10%	11%	9%
Cloud Migration and Infrastructure	16%	13%	12%	14%	14%	10%	12%	16%	11%	19%	11%	18%	17%	6%	19%	13%	14%	11%	15%	17%
Collaboration and Communication	13%	12%	12%	11%	12%	16%	14%	9%	13%	9%	12%	0%	10%	9%	11%	11%	12%	13%	15%	9%
Data Integration and Management	21%	19%	17%	17%	19%	16%	18%	16%	17%	18%	19%	18%	23%	19%	16%	13%	15%	17%	22%	20%
Data Visualization and Reporting	20%	26%	28%	23%	20%	24%	20%	19%	22%	17%	23%	29%	27%	26%	17%	24%	27%	23%	17%	20%
Security and Compliance	5%	2%	3%	3%	3%	7%	4%	9%	9%	6%	3%	0%	0%	4%	6%	8%	4%	4%	7%	3%
Training and Change Management	13%	15%	17%	18%	15%	18%	15%	13%	15%	10%	16%	12%	17%	23%	11%	15%	11%	17%	11%	13%

BI tools enable faster, evidence-based decision processes. The combination of these two drivers suggests that organizations typically view BI not only as an operational tool but also as a strategic asset that supports planning, forecasting, and performance monitoring.

"Data Management and Integration" (16%) is another foundational driver, indicating that organizations view BI as a means to consolidate fragmented data landscapes. Sectors with complex operational structures —such as Agriculture, Forestry and Fishing (I-3) or Manufacturing (I-11)—show particularly high emphasis on this driver, likely due to the need to integrate heterogeneous data sources.

Sector differences are also notable. For example, the Education sector (I-6) has a comparatively strong focus on "Data-Driven Decision Making," reflecting the increasing use of analytics for student outcomes, resource allocation, and institutional planning. Conversely, industries such as Wholesale and Retail Trade (I-20) place greater emphasis on "Customer Experience and Engagement," reflecting their focus on personalized services and insights into customer behavior.

"Environmental and Sustainability Goals" remain low across sectors, although this may reflect the nature of vendor-curated testimonials rather than a lack of overall sustainability activity. As regulatory and ESG pressures increase, this driver may become more visible in future datasets.

Figure 3 provides further insight into the variability of these drivers. "Operational Efficiency" shows a compact distribution with most values concentrated at the upper end, indicating strong consensus across sectors. In contrast, "Data Management and Integration" and "Customer Experience and Engagement" show wider spreads, suggesting more diverse sectoral priorities.

"Cost Efficiency" and "Compliance and Security" generally show lower variability, suggesting a consistent recognition of their importance across sectors, albeit not as critical as operational efficiency. "Innovation and Modernization" appear to have an emphasis across sectors, but with less urgency than operational drivers.

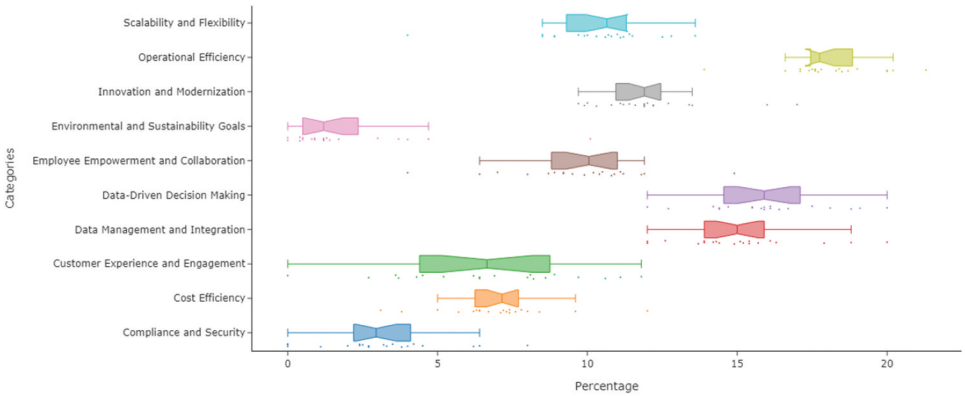


Figure 3. Boxplot of adoption drivers based on industry sectors.

RO2: Implementation strategies

Overall, the implementation strategies across industry sectors (Table 6) point to a shared focus on making BI systems usable in day-to-day organizational work. Within this pattern "Data Visualization and Reporting" is the most commonly referenced strategy (25%), underscoring the centrality of dashboards, reporting tools, and visual displays in contemporary BI deployments. These tools serve as the interface through which organizational users engage with analytics, making their impact both broad and visible.

"Data Integration and Management" (21%) also features prominently. The consistency of this category across sectors indicates that organizations view data consolidation and structured data governance as prerequisites for successful BI deployment. This is particularly evident in sectors such as Public Administration and Defence (I-16), which generate large volumes of structured data requiring alignment across multiple operational domains.

"Cloud Migration and Infrastructure" (9%) reflects the broader transition toward cloud-based BI solutions. Sectors with high digital maturity, such as Information and Communication (I-10), show relatively strong emphasis on cloud adoption, highlighting its perceived value in scalability and availability. However, sectors with sensitive data, such as Financial and Insurance Activities (I-8), demonstrate more varied patterns, likely reflecting stricter regulatory and security requirements.

"Training and Change Management" (15%) consistently appears across sectors, emphasizing that technological implementation alone is insufficient without efforts to support user adoption. Sectors undergoing rapid transformation, such as Education (I-6) and Professional Services (I-15), show higher values, indicating that effective BI use requires engagement, capability building, and cultural adjustment.

The adoption of "AI, Machine Learning, and Advanced Analytics" remains moderate overall (10%) but is notably higher in data-intensive sectors (e.g., I-10). This reflects a gradual shift toward integrating predictive and prescriptive analytics within BI ecosystems, although such capabilities are not yet uniformly embedded across industries.

The boxplot in Figure 4 highlights differences in the variability of strategies. "Data Integration and Management" has a narrow interquartile range and consistently high medians, emphasizing its universal role. Meanwhile, "Cloud Migration and Infrastructure" shows a much broader range, indicating differing levels of cloud readiness and tolerance for security risks across industries.

RO3: Challenges

Challenges identified in the testimonials (Table 7) mirror long-standing issues in BI implementation. "Integration and Interoperability Issues" are the most frequently reported challenge across most sectors, reflecting the difficulty of connecting multiple systems, legacy platforms, and diverse data types. This challenge is particularly evident in sectors such as Wholesale and Retail Trade (I-20), where large-scale transactions, supplier systems, and customer data streams must be aligned.

"Manual Processes and Time-Consuming Tasks" also remain prominent, especially in Public Administration and Defence (I-16) and Real Estate Activities (I-17). These sectors often rely on established procedures and legacy administrative systems, making them susceptible to delays, errors, and inefficiencies when manual workarounds are required.

"Data Quality and Accuracy Issues" (notably high in Mining and Quarrying, I-12) highlight the challenges of ensuring accurate and consistent data in environments

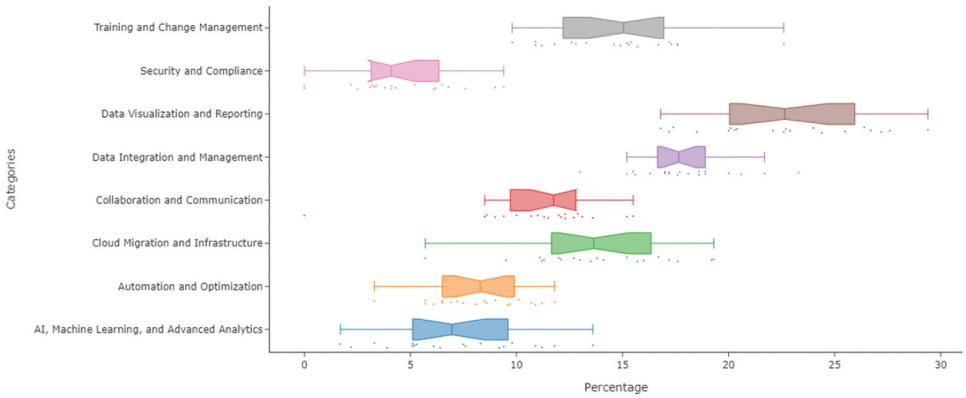


Figure 4. Boxplot of implementation strategies based on industry sectors.

Table 7. Challenges for different sectors.

	Industry Sector*																			
	I-1	I-2	I-3	I-4	I-5	I-6	I-7	I-8	I-9	I-10	I-11	I-12	I-13	I-14	I-15	I-16	I-17	I-18	I-19	I-20
Adoption and Change Management Issues	3%	3%	7%	2%	4%	4%	4%	4%	5%	2%	4%	0%	3%	2%	3%	2%	4%	5%	0%	3%
Data Quality and Accuracy Issues	11%	10%	7%	9%	13%	9%	9%	8%	7%	7%	11%	17%	12%	16%	6%	6%	11%	10%	5%	9%
Data Visualization and Reporting Challenges	14%	18%	16%	17%	14%	21%	14%	12%	16%	13%	16%	11%	18%	21%	13%	16%	16%	17%	10%	13%
Integration and Interoperability Issues	22%	21%	20%	23%	20%	21%	21%	19%	21%	20%	21%	17%	18%	21%	20%	18%	17%	21%	20%	23%
Legacy Systems and Technological Limitations	7%	6%	13%	4%	6%	4%	9%	6%	4%	4%	8%	0%	6%	3%	5%	8%	4%	8%	8%	7%
Manual Processes and Time-Consuming Tasks	18%	19%	18%	19%	20%	19%	21%	15%	17%	15%	21%	17%	21%	19%	16%	25%	24%	17%	15%	17%
Resource and Cost Management Issues	4%	4%	2%	4%	4%	3%	3%	5%	4%	8%	2%	17%	3%	5%	7%	3%	3%	2%	8%	4%
Scalability and Performance Issues	14%	12%	12%	15%	14%	9%	12%	17%	15%	22%	11%	11%	12%	9%	19%	9%	12%	12%	18%	18%
Security, Compliance, and Governance Concerns	4%	2%	0%	2%	3%	4%	4%	8%	9%	6%	2%	6%	3%	0%	6%	6%	4%	4%	8%	3%
Skills and Knowledge Gaps	2%	6%	7%	6%	3%	6%	5%	6%	5%	5%	4%	6%	6%	5%	5%	6%	6%	3%	10%	5%

with complex operational processes. Poor data quality undermines analytics and can diminish trust in BI outputs, reinforcing the importance of effective data governance.

"Scalability and Performance Issues" appear most frequently in sectors dealing with extensive datasets, such as Information and Communication (I-10). These challenges reflect capacity constraints as organizations expand their BI usage, integrate more data sources, or demand faster processing.

Challenges related to "Legacy Systems and Technological Limitations" show how entrenched system architectures can slow or complicate BI adoption. Sectors with longstanding operational infrastructure (e.g., I-7 and I-19) are particularly affected, reinforcing the need for phased modernization.

The boxplot in Figure 5 demonstrates substantial variation in several challenge categories. For example, "Scalability and Performance Issues" shows a wide spread, illustrating that some sectors can scale BI efficiently while others encounter significant bottlenecks. Similarly, variability in "Data Visualization and Reporting Challenges" suggests uneven maturity in translating data into usable insights.

RO4: Collaboration and culture

Collaboration and cultural practices (Table 8) reveal how organizations seek to embed BI into everyday work. "Collaboration and Integration" is consistently the most frequently cited category across nearly all sectors, underscoring the recognition that BI initiatives must break down silos and support information sharing. The uniformity of these values indicates a shared understanding across industries that analytics must be accessible and integrated across teams to generate organizational value.

"Data Accessibility and Self-Service" is particularly prominent in sectors such as Education (I-6) and Other

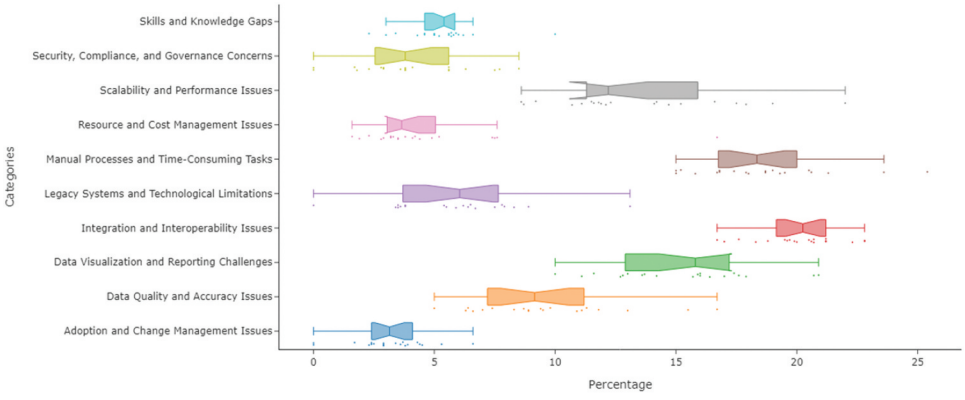


Figure 5. Boxplot of implementation challenges based on industry sectors.

Table 8. Collaboration strategies for different sectors.

	Industry Sector*																			
	I-1	I-2	I-3	I-4	I-5	I-6	I-7	I-8	I-9	I-10	I-11	I-12	I-13	I-14	I-15	I-16	I-17	I-18	I-19	I-20
Automated Workflows and Efficiency	12%	10%	13%	7%	15%	5%	12%	11%	9%	12%	13%	11%	12%	6%	13%	11%	16%	9%	15%	12%
Collaboration and Integration	23%	24%	22%	22%	23%	23%	23%	23%	23%	24%	23%	26%	24%	23%	24%	23%	21%	22%	27%	24%
Cultural Change and Adoption	10%	10%	6%	13%	10%	15%	9%	11%	10%	10%	10%	0%	12%	11%	9%	11%	10%	13%	7%	10%
Customized and Tailored Solutions	11%	8%	10%	12%	10%	11%	11%	9%	10%	12%	8%	21%	6%	13%	12%	7%	11%	12%	12%	10%
Data Accessibility and Self-Service	12%	17%	15%	17%	12%	18%	13%	13%	16%	14%	16%	16%	21%	16%	14%	16%	15%	16%	7%	13%
Real-time Data and Decision Making	17%	18%	18%	16%	16%	14%	15%	16%	17%	19%	15%	16%	12%	13%	17%	17%	17%	18%	20%	18%
Standardization and Governance	12%	9%	10%	8%	9%	9%	11%	12%	12%	7%	11%	11%	9%	11%	8%	12%	7%	8%	10%	9%
Training and Empowerment	1%	4%	5%	5%	4%	5%	6%	5%	5%	3%	5%	0%	3%	8%	3%	3%	4%	3%	2%	4%

Service Activities (I-13). In these settings, BI supports distributed decision-making and empowers users to interact directly with data, reducing reliance on technical teams.

"Real-time Data and Decision Making" is particularly evident in sectors where operational responsiveness is crucial, such as Water Supply and Waste Management (I-19). These organizations depend on up-to-date information to manage system loads, ensure service continuity, and identify anomalies quickly.

"Standardization and Governance" is given greater emphasis in regulated sectors, including Public Administration and Defence (I-16). These sectors require strong processes for data stewardship, auditability, and compliance.

The boxplot in Figure 6 indicates that while collaboration-oriented categories are uniformly valued, other categories—such as "Cultural Change and Adoption"—show greater variation. This reflects differing organizational readiness levels; some sectors have embedded data-driven cultures, while others remain early in their analytics maturity journey.

These findings illustrate how the drivers, implementation strategies, challenges, and cultural aspects of BI adoption vary across sectors while also revealing several consistent patterns. The following discussion examines these relationships, highlights their interconnections, and considers their organizational implications.

Discussion

This section examines implications, interrelationships, and industry-specific nuances, emerging from the findings. By synthesizing these findings, we aim to provide a comprehensive understanding and actionable recommendations.

Correlation analysis

Correlation analysis was used to explore associations among the categories identified in the study. These correlations highlight notable patterns but should not be interpreted as causal relationships.

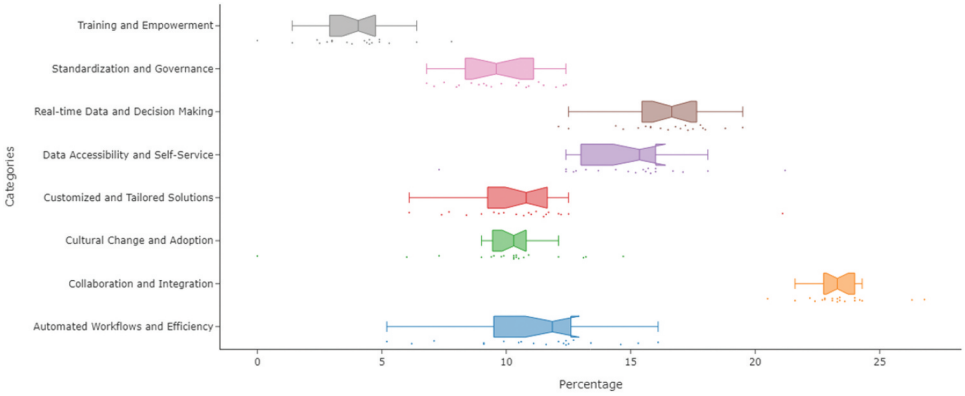


Figure 6. Boxplot of collaboration strategies based on industry sectors.

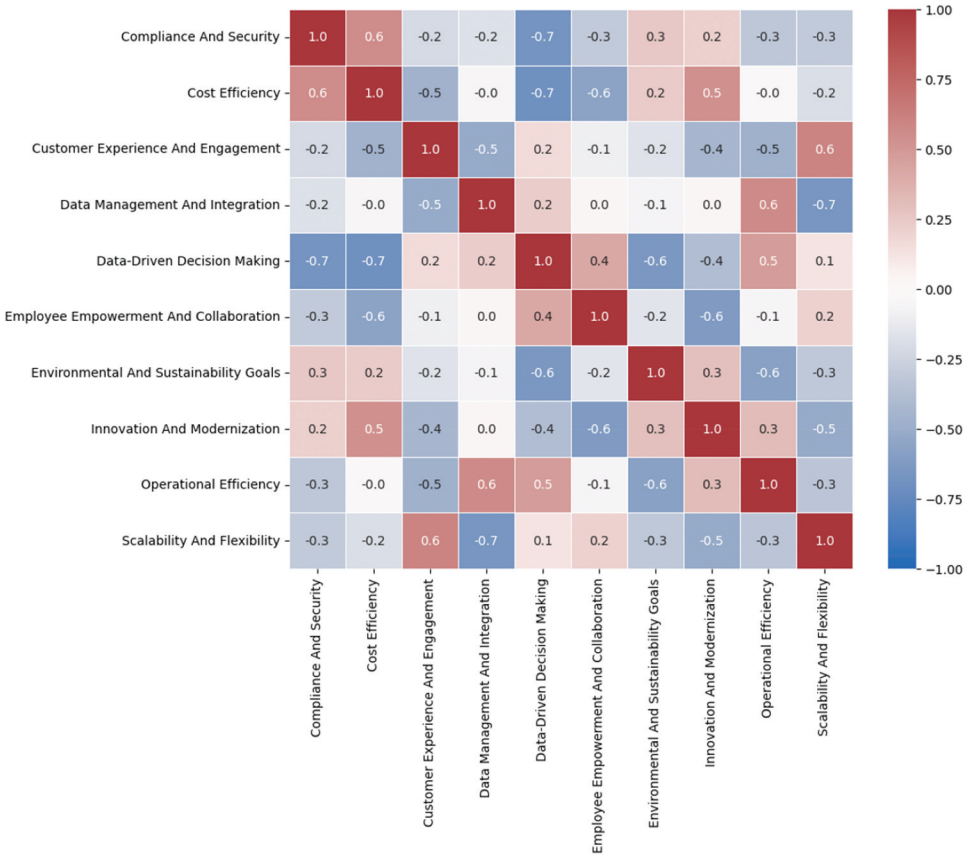


Figure 7. Correlation matrix for adoption drivers.

Adoption drivers

The correlation matrix (Figure 7) indicates that *Operational Efficiency* and *Data-Driven Decision Making* frequently co-occur (0.5), suggesting that organizations often pursue BI to achieve process improvements while strengthening analytical capabilities. *Data Management and Integration* also shows a moderate positive relationship with *Operational Efficiency* (0.6), reinforcing the centrality of data infrastructure to operational gains.

Some correlations highlight potential tensions. The strong negative association between *Data-Driven Decision Making* and *Cost Efficiency* (−0.7) suggests that aspirations for advanced analytics may be challenging to reconcile with tight budgets. A similar trade-off appears between *Scalability and Flexibility* and *Data Management and Integration* (−0.7), suggesting that organizations prioritizing adaptable BI solutions may be less able to invest in extensive integration work.

Implementation strategies

The matrix in Figure 8 reveals several meaningful interrelationships. A positive correlation (0.5) between "*Training and Change Management*" and "*Data Visualization and Reporting*" indicates that user capability is closely tied to the effective use of BI reporting tools. This highlights the continued importance of investing in user readiness.

A strong negative correlation between *Cloud Migration and Infrastructure* and *Training and Change Management* (−0.8) suggests that organizations adopting cloud-based BI tools may perceive them as easier to use, potentially reducing formal training efforts. The negative association between *AI, Machine Learning, and Advanced Analytics* and *Collaboration and Communication* (−0.6) may point to challenges in translating advanced analytical outputs into a shared organizational understanding.

Implementation challenges

The challenges matrix (Figure 9) highlights how certain difficulties tend to reinforce one another. The positive association between *Manual Processes and Time-Consuming Tasks* and *Data Visualization and Reporting Challenges* suggests that reliance on manual data preparation undermines the production of timely and high-quality reporting.

The strong negative correlation between *Adoption and Change Management Issues* and *Resource and Cost*

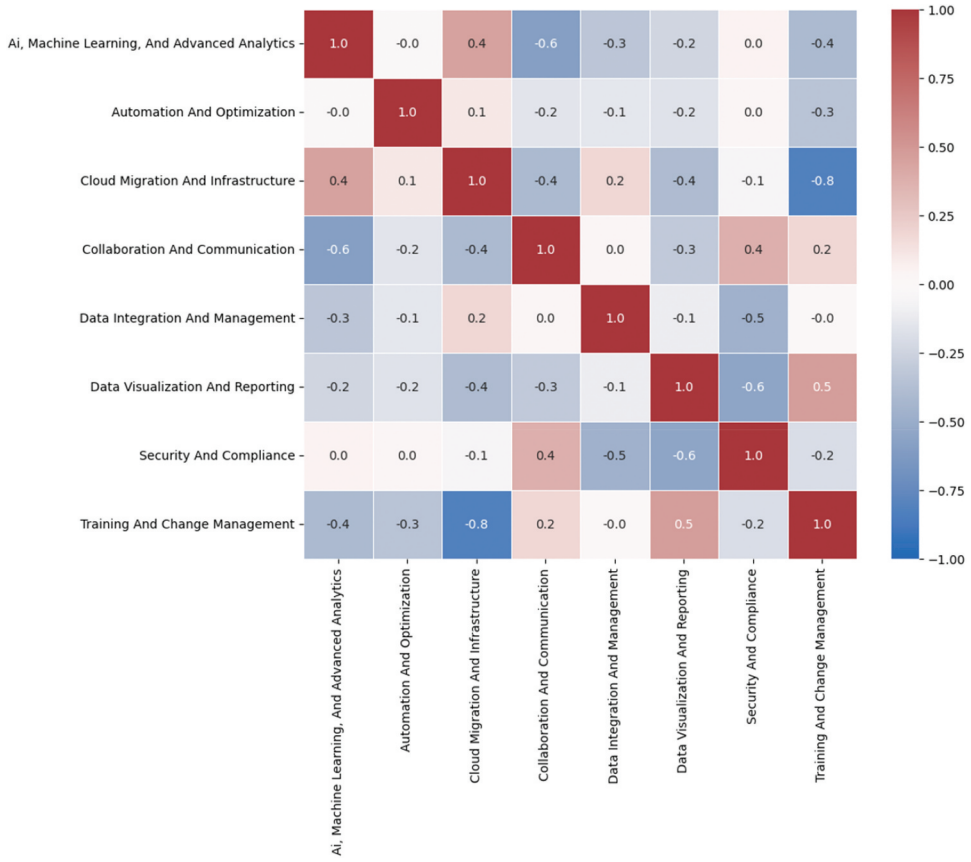


Figure 8. Correlation matrix for implementation strategies.

Management Issues (−0.7) indicates that organizations investing resources in structured change processes may experience fewer adoption difficulties, whereas those with limited capacity may struggle to support users effectively.

Collaboration and culture

The correlation matrix in Figure 10 shows that *Training and Empowerment* is positively associated with *Cultural Change and Adoption* (0.4), reflecting the role of training in building confidence and fostering a data-driven culture.

Data Accessibility and Self-Service displays a moderate negative correlation with *Automated Workflows and Efficiency* (−0.5), suggesting a possible trade-off between user-led exploration and centralized automation. By contrast, *Real-Time Data and Decision Making* shows a positive correlation with *Automated Workflows and Efficiency* (0.4), indicating that real-time data capability strengthens operational responsiveness. Its negative association with *Data Accessibility and Self-Service* (−0.6) may reflect the technical complexity and continuous updating required for real-time systems.

Strategic insights

Drawing on the empirical patterns observed across sectors, the analysis offers several implications for organizations planning or refining BI initiatives. First, the interlinked nature of adoption drivers suggests that BI investments produce greater value when approached holistically. Efforts aimed solely at operational improvement or analytic capability may be less effective than integrated strategies recognizing their mutual reinforcement.

Second, the findings highlight the foundational importance of data management. High-quality, integrated data environments underpin both operational gains and reliable decision-making. Organizations should therefore prioritize data architecture, governance, and integration as prerequisites for more advanced BI activities.

Third, the correlations point to clear trade-offs. Ambitions for scalable or flexible BI solutions, or for advanced analytics, often conflict with cost constraints. Organizations may need to evaluate which capabilities are essential and phase development accordingly.

Fourth, the prominence of visualization and reporting strategies emphasizes the need for BI tools that

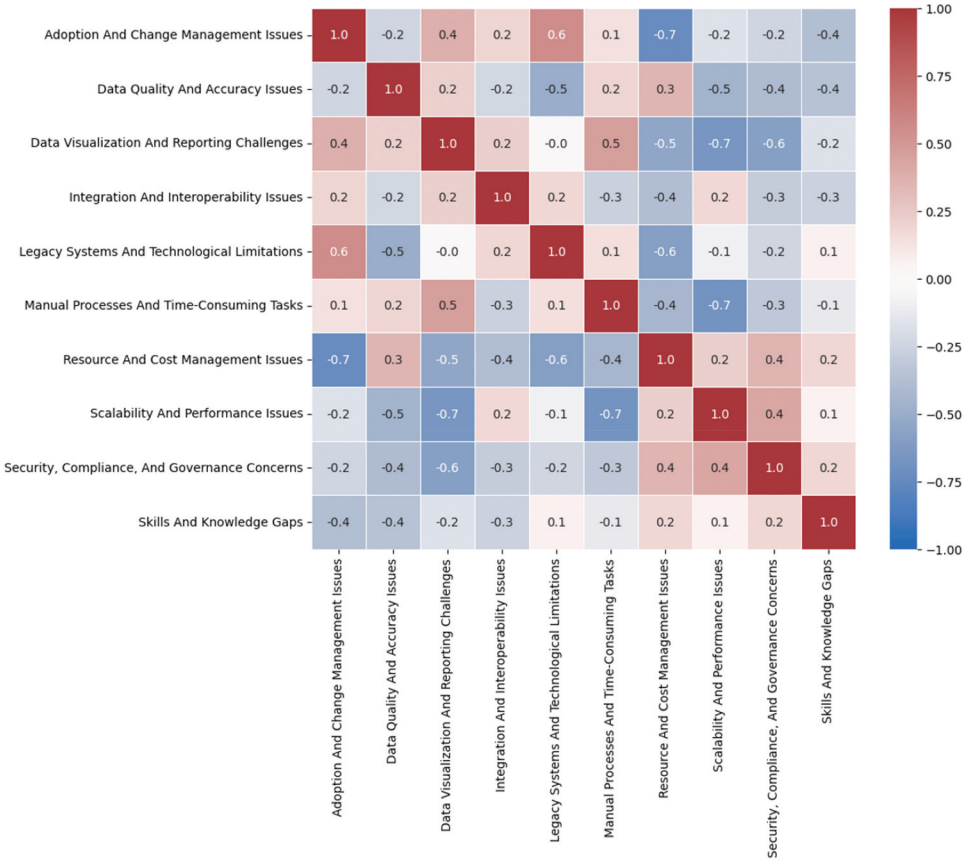


Figure 9. Correlation matrix for implementation challenges.

communicate insights clearly. The emerging role of cloud solutions and advanced analytics suggests that organizations should assess their data workloads, regulatory environment, and infrastructure maturity when selecting BI architectures.

Fifth, organizations relying heavily on manual data processes face compounded challenges in reporting and efficiency. Investing in automation and modern integration pipelines can significantly reduce these burdens and strengthen BI outcomes.

Finally, the analysis reinforces the critical role of people and organizational culture. Training, user enablement, and structured change management are closely linked with the successful uptake of BI tools. Organizations seeking to build a data-driven culture should therefore combine technical enhancements with deliberate efforts to develop user capability.

Conclusions

This study applies an LLM-based approach to analyze approximately 2,800 customer testimonials, addressing the limited empirical use of large-scale narrative data in BI research. Rather than modeling adoption behavior, the analysis identifies recurring themes in real-world BI experiences using GPT-4 to structure and interpret unstructured customer accounts. The findings highlight organizations' key motivations-operational efficiency, data-driven decision-making, and investment in robust data infrastructure-alongside growing interest in cloud-based and AI-enhanced BI capabilities. Persistent challenges include data integration difficulties, reliance on manual processes, and skills gaps, underscoring the importance of training and the development of a data-driven culture.

The study also contributes to understanding how BI experiences are presented within industry narratives. Vendor-curated testimonials naturally emphasize positive outcomes and underrepresent unsuccessful or neutral cases, introducing a dataset bias. Future studies should incorporate a wider range of narrative sources to provide a more balanced assessment of BI adoption.

A further limitation concerns the use of GPT-4 itself. LLMs are trained on extensive web-scraped material, which may include marketing content similar to the testimonials analyzed. As a result, the model may reflect patterns that favor positive framings of technology

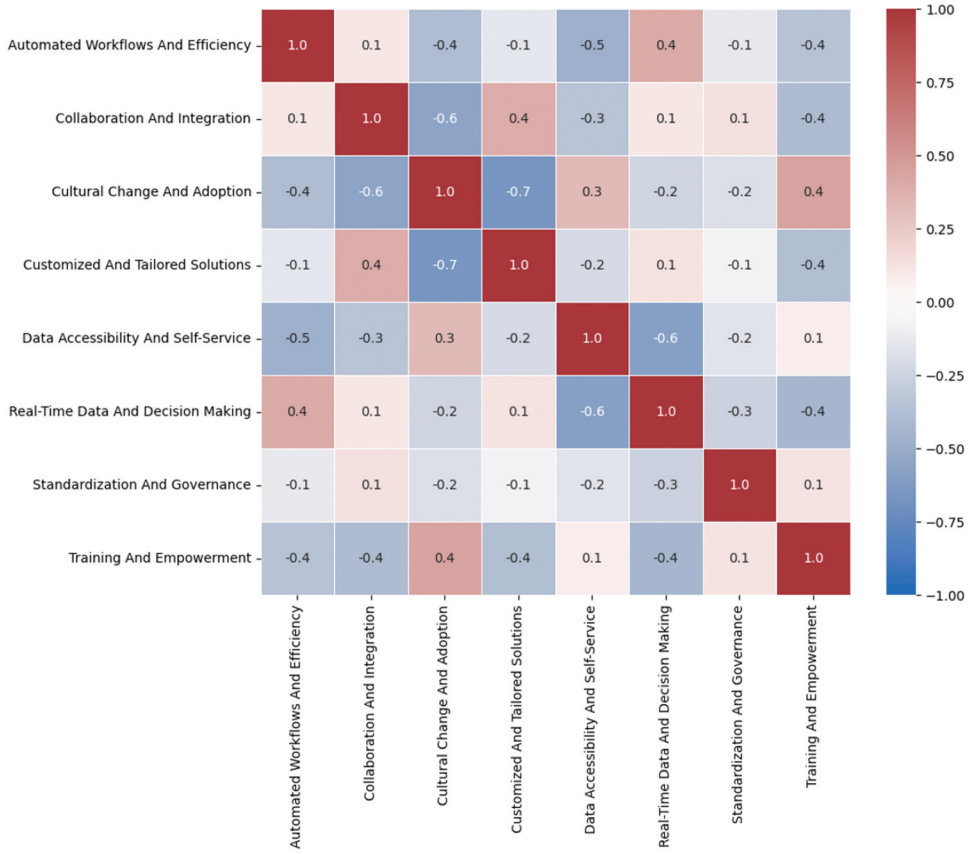


Figure 10. Correlation matrix for collaboration and culture.

adoption, potentially influencing theme extraction. The use of LLMs as analytical tools also raises methodological considerations, as their interpretive processes differ from human reasoning and are not directly observable. While the validation procedures mitigate immediate reliability concerns, the possibility of bias in the training data remains an important consideration in qualitative research using LLMs.

The application of UK SIC codes provided structure for industry comparison but may not fully capture the nuances of emerging or hybrid sectors, potentially limiting generalizability. In addition, although GPT-4 supports scalable analysis, it may not always capture contextual or sector-specific subtleties within complex customer accounts.

The relationships identified in this study should be interpreted as indicative rather than causal, reflecting the narrative and descriptive nature of the data. Given the pace of technological development in BI, the patterns observed may also evolve over time.

Future research could examine the longer-term effects of BI adoption, including operational and financial outcomes, and explore how different user groups engage with BI tools. Combining narrative data with quantitative performance indicators may offer a more comprehensive understanding of BI success and its organizational impact.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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